

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)****Y25 - 2026-2027 Trimester - 4****Project Abstract Submission Form**

Sec No- 8	Team No- 9	Project Title- Online Banking System Simulation
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1. Abstract:

The Online Banking System Simulation is a backend-focused project that models core banking operations such as account management, deposits, withdrawals, fund transfers, and transaction history retrieval. The system is designed as a layered backend service, separating request handling, validation, business logic, data access, and persistence, so that a single request can be traced from HTTP arrival to JSON response. The underlying relational schema is designed using entity-relationship modelling and normalized up to BCNF to remove redundancy and update anomalies while preserving referential integrity across customers, accounts, and transactions. SQL is used extensively for schema definition, data manipulation, joins, aggregations, subqueries, CTEs, and window functions to generate statements and reports.

Since banking operations are transaction-critical, the system enforces ACID properties, appropriate isolation levels, and safeguards against concurrency anomalies such as dirty reads and phantom reads, ensuring balances stay consistent under concurrent access. Query performance is optimized using B-tree and covering indexes along with EXPLAIN plans to identify and eliminate slow queries. To simulate real-world scalability, the project incorporates distributed database concepts such as primary-replica replication for read scaling and horizontal sharding by customer ID, reasoning about trade-offs using the CAP theorem. The final deliverable is a working simulation demonstrating reliable, consistent, and scalable banking transactions backed by a well-engineered relational and distributed database layer.

2. Team Details:

Roll Number(s)	Name of the Student	Signature
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3. Frameworks / Platforms / Software Used:

- Backend: Node.js with Express.js (REST API layer, service/business logic)
- Database: PostgreSQL (relational schema, transactions, indexing, replication)
- Database Tools: pgAdmin / MySQL Workbench for schema design and EXPLAIN plan analysis
- API Testing: Postman
- Version Control: Git and GitHub
- Containerization / Deployment: Docker (for simulating multi-node / replica setups)
- Language: JavaScript / TypeScript

Guide Name and Signature _____